

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Multi-Turn DPO for Tool-Augmented LLMs

5-State MDP · 3 Query Types · Automatic Preference Pairs

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Three Failure Modes of SFT-Only Training:

1

Failing to Ask Follow-Up Questions

TA-LLMs with missing arguments do not ask for clarification, leading to hallucinated tool calls.

2

Making Hallucinated Tool Invocations

Models invoke tools with missing or incorrect arguments when information is incomplete.

3

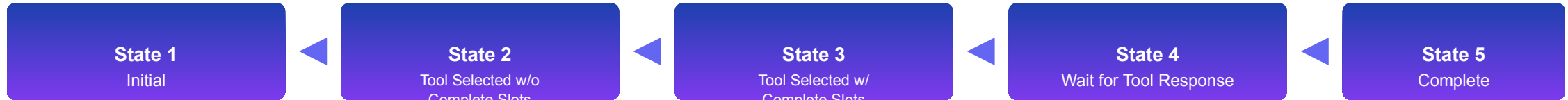
Failing to Reject Inappropriate Tool Calls

Out-of-scope requests are not rejected, leading to invalid tool usage.

Key Insight: SFT alone has limitations in learning conversational skills needed for tool-augmented dialogue.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

5 Internal States (Unobserved):



State Descriptions:

State 1 — Initial

No history. Out-of-scope requests return rejection.

State 2 — Tool Selected w/o Slots

Tool identified, parameters missing. Slot-filling QA occurs here

State 3 — Tool Selected w/ Slots

Tool selection and all argument values determined.

State 4 — Wait for Tool Response

Tool call executed, awaiting results.

State 5 — Complete

Tool results received, completion message generated.

Key Insight:

The 5-state MDP formulation models the complete dialogue flow for tool-augmented LLMs, enabling systematic construction of correct and incorrect trajectories for preference optimization.

Three Query Types Define Distinct State Trajectories

Query Type Classification Based on State Transition Trajectories:

Query Type	State Trajectory	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ (Complete Info)	Query contains all required arguments. No slot-filling needed.
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$ (Incomplete Info)	Query lacks arguments. Slot-filling QA repeats N times until all fields gathered.
Type 3	$1 \rightarrow 1$ (Out-of-Scope)	Requested functionality unavailable. Tool call must be rejected.

Key Insight: Each query type follows a distinct state trajectory, enabling automatic construction of chosen vs. rejected trajectory pairs for DPO training.

State 2 Loop (2×N)

Type 2 loops in State 2 for each missing slot until all are filled.

State 1 Rejection

Type 3 stays in State 1 when request is out-of-scope.

Trajectory Pairing

Chosen (correct) vs. Rejected (incorrect) trajectories per query.

Automatic Paired Trajectory Construction Without Human Labels

Three Learning Lessons from Paired Trajectory Dataset:

Lesson 1

Prevent Redundant Slot-Filling

Chosen: 1→3→4→5
vs. Rejected: 1→(2*N)→3→4→5

Lesson 2

Prevent Slot Hallucination

Chosen: 1→(2*N)→3→4→5
vs. Rejected: 1→3→4→5

Lesson 3

Tool Call Reject/Accept

Accept: 1→3→4→5 vs. Reject: 1→1
or Reject: 1→(2*N)→3→4

Dataset Composition (FunctionChat-Bench):

Query	Chosen traj.	Rejected traj.	Learning lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

Key Insight: Automatically constructed chosen vs. rejected trajectory pairs — no human labeling required.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

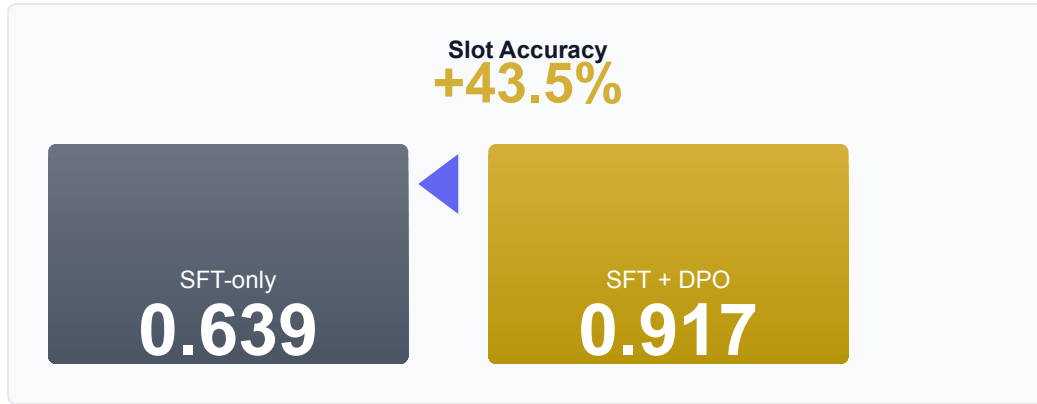
FunctionChat-Bench Results — Call | Completion | Slot | Relevance | Micro Avg | Macro Avg

Model	Method	Call	Compl.	Slot	Relev.	Micro	Macro
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

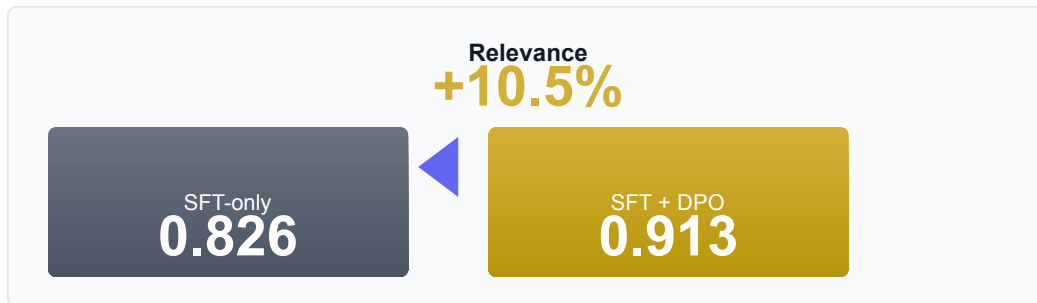
Key Insight: SFT + DiaTool-DPO consistently achieves the highest Macro Avg across all three model sizes.
Near GPT-4o performance: 94.8% info gathering, 91% tool call rejection.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Slot Accuracy (SFT-only → SFT + DiaTool-DPO):



Relevance (SFT-only → SFT + DiaTool-DPO):



Why These Gains Matter

Slot Accuracy: 0.639 → 0.917

Prevents hallucinated tool invocations by correctly filling missing slots before making a tool call.

Relevance: 0.826 → 0.913

Improves tool call rejection accuracy for out-of-scope requests, reducing invalid tool usage.

Micro Avg: 0.844 → 0.905

Overall dialogue quality uplift from combined SFT + DPO approach.

Macro Avg: 0.816 → 0.904

Best balanced performance across all metrics, approaching GPT-4o level.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1

MDP Formulation

5 internal states and 3 query types enable systematic trajectory pair construction without human annotation.

2

Automatic Dataset

Chosen vs. rejected trajectory pairs are automatically constructed — no human labeling required.

3

DPO Complements SFT

DPO alone is insufficient; DPO enhances SFT for dialogue control — not a replacement.

4

Language-Agnostic

Primary experiments in Korean, but method generalizes to English, opening broad applicability for production TA-LLMs.

5

Path to GPT-4o-Level

Enables near GPT-4o conversational tool use on smaller open-source models (Prop.-8B, Prop.-3.1B, LLaMA-3-8B).



Best Macro Average

SFT + DiaTool-DPO achieves highest Macro Avg across all model sizes, with +43.5% Slot and +10.5% Relevance.

Research Impact

- DiaTool-DPO formulates tool-augmented dialogue as a 5-state MDP, enabling systematic preference optimization
- Automatic dataset construction eliminates expensive human labeling while maintaining quality
- DPO complements SFT for dialogue control — approach is language-agnostic and model-agnostic
- Opens path to GPT-4o-level conversational tool use on smaller production models

Paper: [arXiv:2504.02882](https://arxiv.org/abs/2504.02882) | Near GPT-4o performance: 94.8% info gathering, 91% tool call rejection